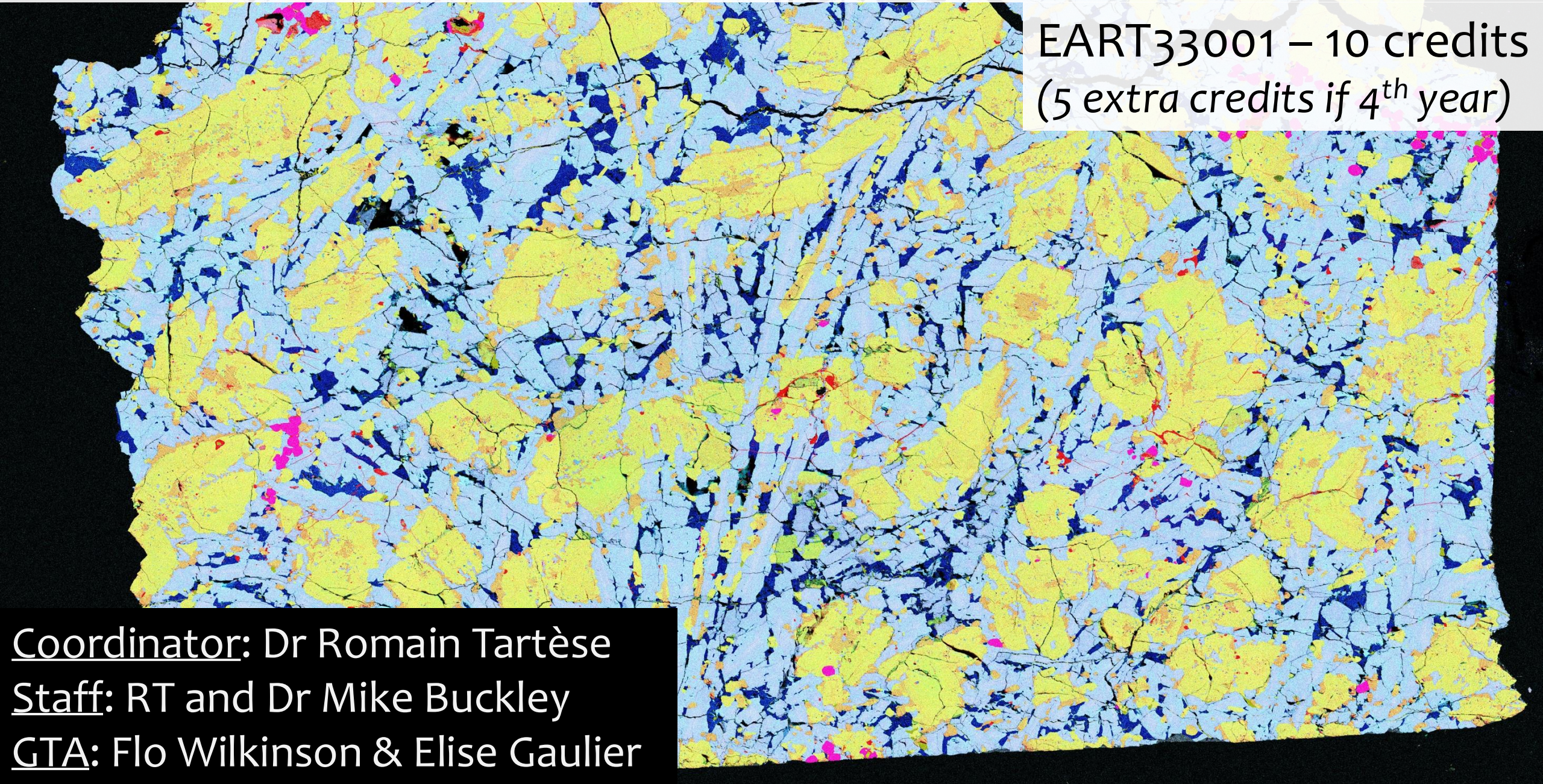
Analytical Techniques in Earth and Environmental Science



EART33001 – 10 credits
(5 extra credits if 4th year)

<u>Coordinator</u>: Dr Romain Tartèse
<u>Staff</u>: RT and Dr Mike Buckley
<u>GTA</u>: Flo Wilkinson & Elise Gaulier

Student Voice

What students told us last year:

- “Gives good overview of various methods of analysis, weeks feel like they follow on from one another rather than being unrelated, despite addressing a broad range of topics. Practicals seem thorough and useful for revision for Jan 26”.
- “Assessments were clear and well structured”; “Minimal course load, well organized content”.
- “have practicals that related to the apparatus introduced rather than calculation questions”.
- “I hope there are after–class videos for review”.
- “Weekly ILOs as well as overall unit ones”.

How we have changed it for this year:

- Lots of positives, which is always nice.
- Revised some of the practicals to better match technique described.
- All lectures are podcasted and available to watch afterwards (they already were).
- The course is built as a whole progression throughout the semester to achieve the ILOs. As such, weekly ILOs would not make a lot of sense.

Your Voice Matters

How you can give feedback

Speak to me!

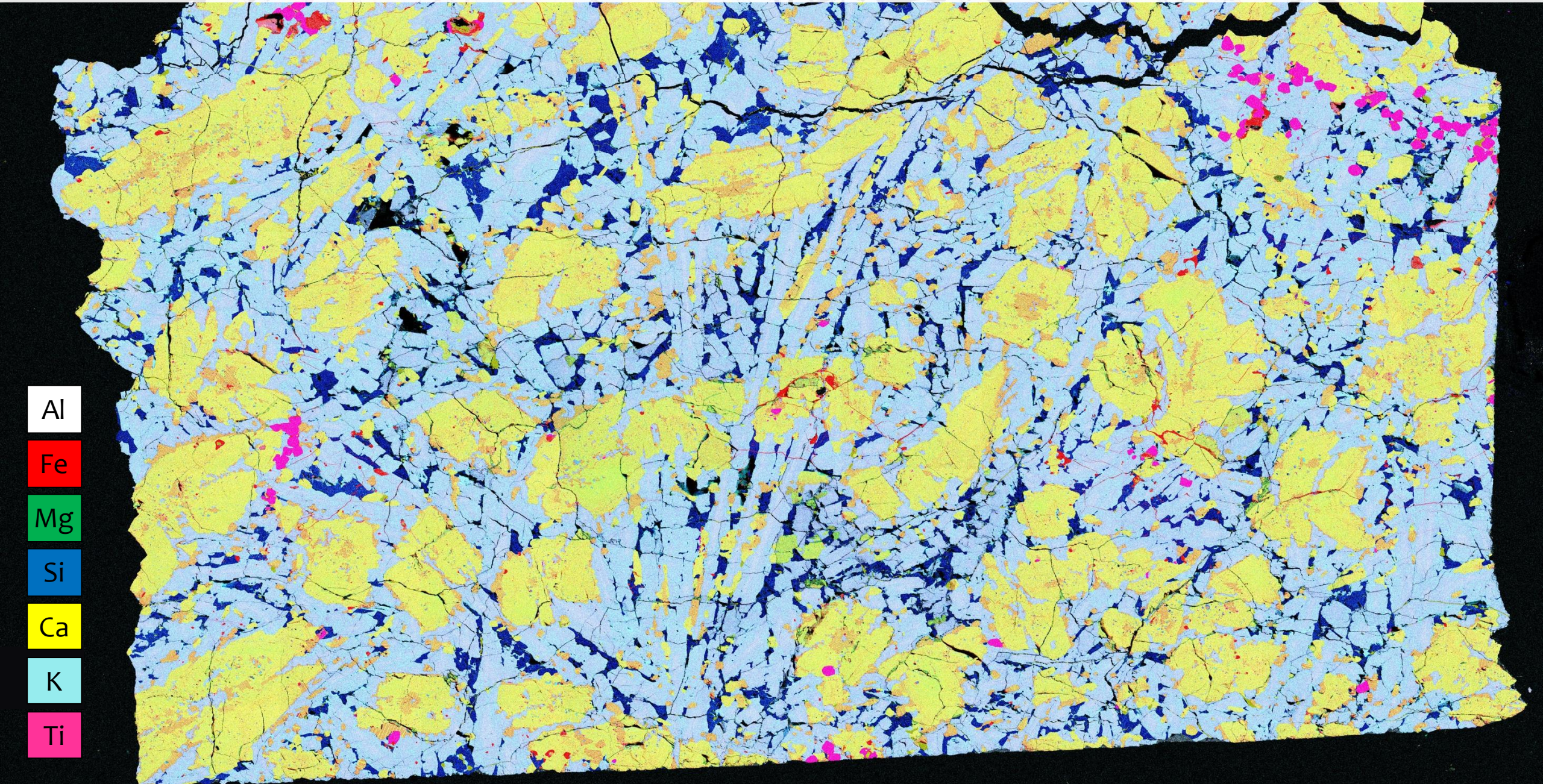
Contact your Student Representatives

Complete surveys

Attend events like Faculty Forum

Find out how to make your voice heard

Erg Chech 002 – oldest magmatic meteorite



Lab instruments are nothing more than tools!

Sophisticated, complex, but still tools.
Understanding their principles, and how they works, lead to better use, possibly new applications, and sometimes new science questions.

©D. Descouens / Wikipedia



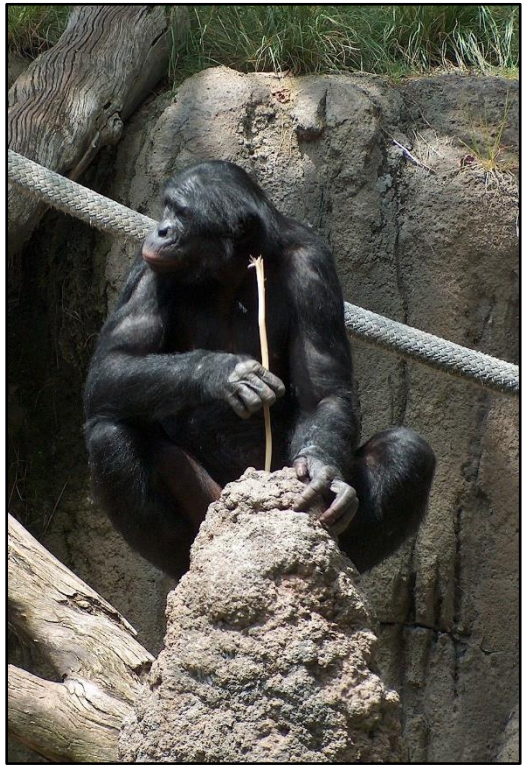
Oldowan tools – sharp stones
used since at least 2.6 Ma.

©O.S. Muhammed Amin / Wikipedia

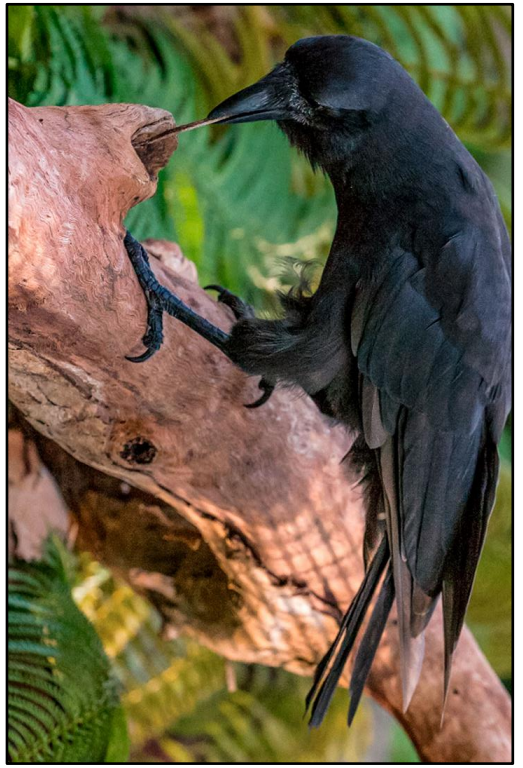


Acheulean – hand axes
used since at least 1.8 Ma.

©Mike R / Wikipedia



©Ken Bohn / San Diego Zoo Global



Why are we teaching this course?

"The aim of this course is to teach students the **fundamental physical and chemical principles** that underpin modern analytical methods in Earth and Environmental Science research. This knowledge is fundamental to a career as a practicing scientist in these fields".







Why are we teaching this course?

For example, typical responsibilities of Environmental scientists, according to targetjobs' Environmental Scientist description, include:

- ***deciding*** *on the best data collection methods*
- ***conducting*** *field surveys and collecting data*
- ***conducting*** *lab tests on water, air and soil samples*
- ***interpreting*** *data to identify whether contamination exists in accordance with environmental laws*
- ***building*** *conceptual models that identify the potential contaminant sources that could have an adverse impact on the environment*
- ***preparing*** *detailed scientific reports or presentations based on their findings*
- ***communicating*** *the results of their studies to senior scientists and key stakeholders*
- ***devising*** *plans to minimise or fix environmental problems*
- *when more senior,* ***taking on*** *project and budget management*

So, by the end of this course you should be able to:

- **Describe** a range of analytical methods used in Earth and Environmental Sciences.
- **Summarise** some cutting-edge developments in analytical methodology.
- **Solve** basic calculations that relate key variables (energy, wavelength) of electromagnetic radiations.
- **Illustrate** how some of the methods studied might be adapted to new applications.
- **Design** ad-hoc research protocols using the knowledge gained during the course.

Transferable skills along the way

EART33001: Analytical Techniques in Earth and Environmental Science

In this unit you will develop the following transferable skills:

Analytical and critical thinking

- Numeracy skills.
- Assessing data quality and uncertainty.
- Conceptualising across a range of spatial and temporal scales
- Manipulating and interpreting data using appropriate numerical and statistical techniques.

Problem-solving

- Developing and evaluating solutions to complex problems.
- Developing solutions by integrating information from diverse disciplines and data types.

Research and learning

- Critically reviewing and evaluating different sources of information and data.
- Learning independently.

Decision-making

- Using different tools to inform decision-making.
- Drawing conclusions with uncertainties from ambiguous datasets.

Wider context awareness

- Applying the subject matter to understanding global issues.

Fieldwork

- Using field measurement techniques and technologies.
- Designing investigations and applying practical techniques regularly used in applied professions.

Laboratory

- Systematic recording, visualisation and interpretation of laboratory data.
- Assessing health and safety risks and applying suitable mitigating measures
- Designing investigations and applying practical techniques regularly used in applied professions.

Digital

- Office suite (Word, Excel, etc.)
- Data visualisation.

Project management

- Working independently.

Interpersonal

- Working cooperatively.

Creativity

- Independently identifying alternative ways to view and solve a problem.

How is the course organised?

Each week, one 50-min ‘lecture’ slot, followed by practical for 2 hours.

- Dedicated Canvas module for each week where I’ll upload relevant material and lecture notes ~1 week in advance.

→ *You should review this material on Canvas in advance of the classes, then revisit it afterwards. Classes provide an opportunity to discuss the material, request further explanation, or bring up ideas for discussion.*

- During practicals, we will work with processing and interpreting datasets relevant to the technique(s) of the week.

→ *Practical sheet model answers posted to relevant Canvas page straight after the practical.*

How is the course organised?

R. Tartèse
M. Buckley

Lecture slot: Fridays 09:00-09:50 (until w/c 09/11) then Mondays 1300-1350 – in Dover Street building room 1.036
Practical slot: Fridays 10:00-11:50 (until w/c 09/11) then Mondays 1400-1550 – in Williamson building room 2.45

Week 2 – X-ray Diffraction (XRD)	Week 7 – Proteomics and Genomics
Week 3 – ICP-OES/ICP-MS	Week 8 – X-ray Computed Tomography (XCT)
Week 4 – Scanning Electron Microscopy and Electron Probe MicroAnalysis	Week 9 – X-ray Absorption Spectroscopy (XAS)
Week 5 – Analysis of solids by mass spectrometry	Week 10 – Raman and FTIR Spectroscopy
Week 6 – Liquid and Gas Chromatography-Mass Spectrometry	Week 11 – Review session (*if needed*)

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In class test: on 30th Oct 2026 from 1300-1500 (50% final mark)

Final exam: during exam period, 1.5 hours (50% final mark)

Who am I, and why am I teaching this course?

Contents lists available at ScienceDirect

Comptes Rendus Geoscience

www.sciencedirect.com

Petrology, geochemistry

New U-Pb zircon and $^{40}Ar/^{39}Ar$ muscovite age constraints on the emplacement of the Lizio syn-tectonic granite (Armorican Massif, France)

Nouveaux âges U-Pb sur zircon et $^{40}Ar/^{39}Ar$ sur muscovite de mise en place du granite syn-tectonique de Lizio (Massif Armoricain, France)

Romain Tartèse [a,*], Marc Poujol [a], Gilles Ruffet [a,b], Philippe Boulvais [a], Philippe Yamato [a], Jan Košler [c]

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

www.elsevier.com/locate/epsl

Lunar basalt chronology, mantle differentiation and implications for determining the age of the Moon

Joshua F. Snape [a,*], Alexander A. Nemchin [a,b], Jeremy J. Bellucci [a], Martin J. Whitehouse [a,c], Romain Tartèse [d,e], Jessica J. Barnes [e], Mahesh Anand [e,f], Ian A. Crawford [g], Katherine H. Joy [h]

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Microfossils with tail-like structures in the 3.4 Gyr old Strelley Pool Formation

Frédéric Delarue [a,*], Sylvain Bernard [b], Kenichiro Sugitani [c], François Robert [b], Romain Tartèse [d], Sonja-Verena Albers [e,f], Rémi Duhamel [b], Sylvain Pont [b], Sylvie Derenne [a]

Geochemical Perspectives Letters

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Warm Archean oceans reconstructed from oxygen isotope composition of early-life remnants

R. Tartèse*[1,2], M. Chaussidon[3], A. Gurenko[4], F. Delarue[1], F. Robert[1]

RESEARCH

LUNAR GEOLOGY

Age and composition of young basalts on the Moon, measured from samples returned by Chang'e-5

Xiaochao Che[1], Alexander Nemchin[1,2]*, Dunyi Liu[1,3]*, Tao Long[1], Chen Wang[1], Marc D. Norman[4], Katherine H. Joy[5], Romain Tartese[5], James Head[6], Bradley Jolliff[7], Joshua F. Snape[5], Clive R. Neal[8], Martin J. Whitehouse[9], Carolyn Crow[10], Gretchen Benedix[2,11], Fred Jourdan[2], Zhiqing Yang[1], Chun Yang[1], Jianhui Liu[1], Shiwen Xie[1], Zemin Bao[1], Runlong Fan[1], Dapeng Li[3], Zengsheng Li[3], Stuart G. Webb[8]

American Mineralogist, Volume 98, pages 1477–1486, 2013

NanoSIMS Pb/Pb dating of tranquillityite in high-Ti lunar basalts: Implications for the chronology of high-Ti volcanism on the Moon

ROMAIN TARTÈSE,[1,*] MAHESH ANAND,[1,2] AND THOMAS DELHAYE[3]

The abundance, distribution, and isotopic composition of Hydrogen in the Moon as revealed by basaltic lunar samples: Implications for the volatile inventory of the Moon

Romain Tartèse [a,*], Mahesh Anand [a,b], Jessica J. Barnes [a,b], Natalie A. Starkey [a], Ian A. Franchi [a], Yuji Sano [c]

Geochronology, 4, 683–690, 2022
https://doi.org/10.5194/gchron-4-683-2022
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GEOCHRONOLOGY Open Access EGU

In situ U–Pb dating of 4 billion-year-old carbonates in the martian meteorite Allan Hills 84001

Romain Tartèse and Ian C. Lyon

Contents lists available at ScienceDirect

Comptes Rendus Geoscience

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Petrology, geochemistry

Geochemical Perspectives Letters

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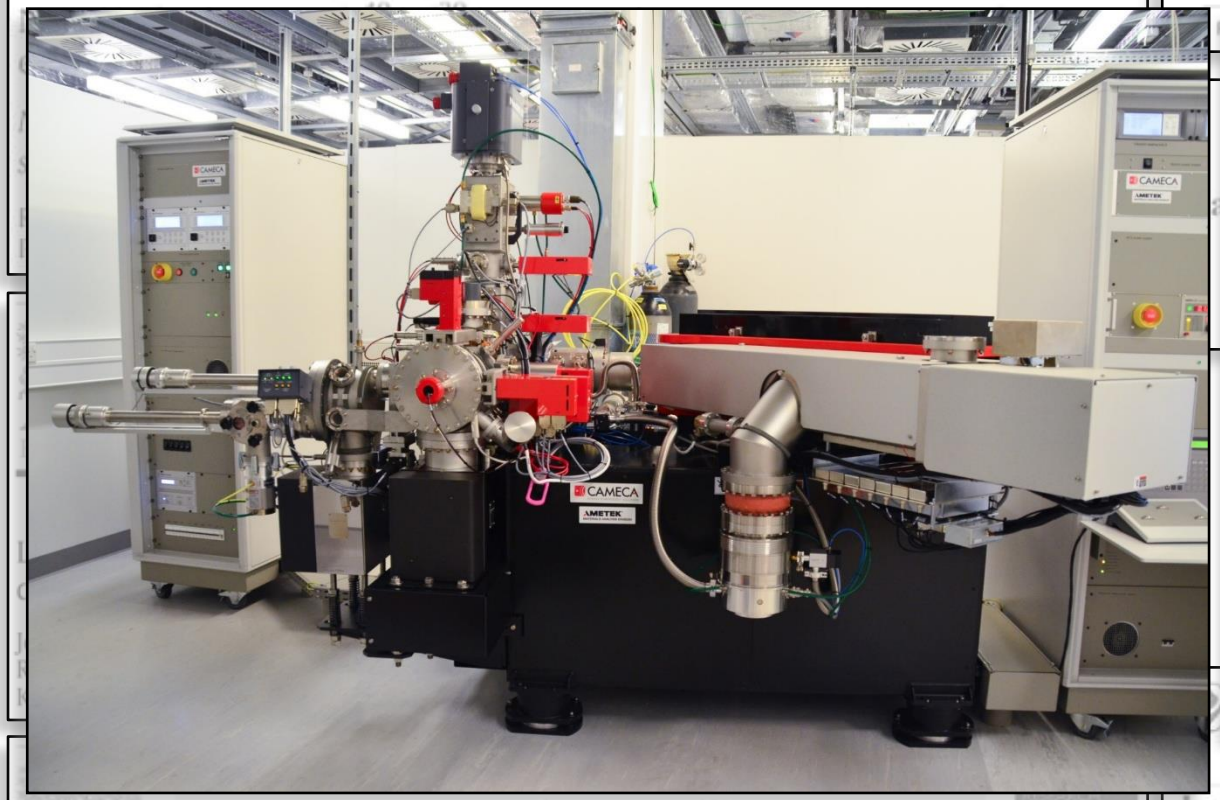
Warm Archean oceans reconstructed from oxygen isotope composition of early-life remnants

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Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Microfossils with tail-like structures in the 3.4 Gyr old Strelley Pool Formation

Frédéric Delarue[a,*], Sylvain Bernard[b], Kenichiro Sugitani[c], François Robert[b], Romain Tartèse[d], Sonja-Verena Albers[e,f], Rémi Duhamel[b], Sylvain Pont[b], Sylvie Derenne[a]

JGR

Journal of Geophysical Research: Planets

RESEARCH ARTICLE

10.1002/2016JE005143

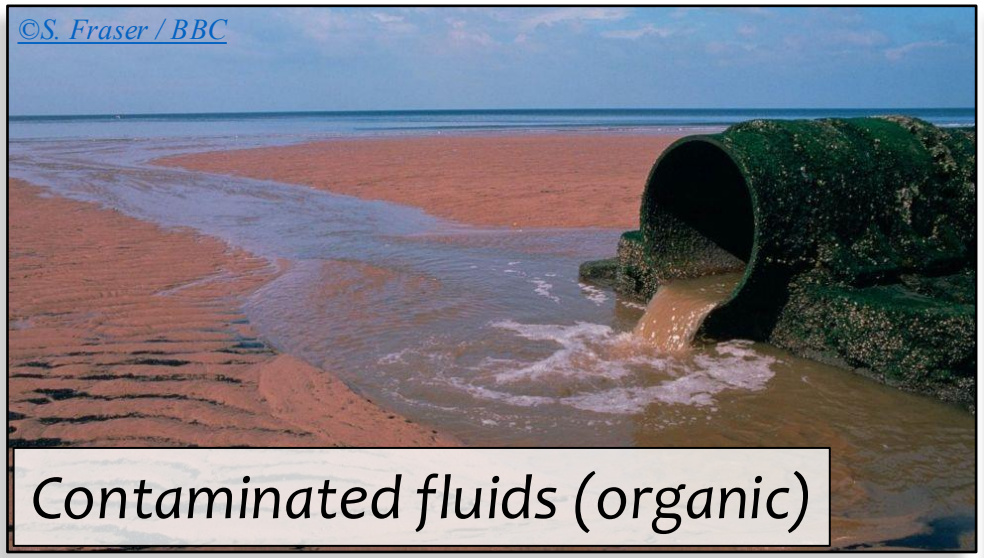
Key Points:

- The Martian breccia NWA 7034 is an air fall deposit that was likely formed by impact processes on Mars
- The earliest crust on Mars formed through alkali-rich volcanism under oxidizing conditions
- Regolith formation processes on Mars

Geologic history of Martian regolith breccia Northwest Africa 7034: Evidence for hydrothermal activity and lithologic diversity in the Martian crust

Francis M. McCubbin[1,2], Jeremy W. Boyce[2,3], Tímea Novák-Szabó[4,5], Alison R. Santos[2], Romain Tartèse[6,7], Nele Muttik[2], Gabor Domokos[4], Jorge Vazquez[8,9], Lindsay P. Keller[1], Desmond E. Moser[10], Douglas J. Jerolmack[5], Charles K. Shearer[2], Andrew Steele[11], Stephen M. Elardo[11], Zia Rahman[12], Mahesh Anand[7,13], Thomas Delhaye[14], and Carl B. Agee[2]

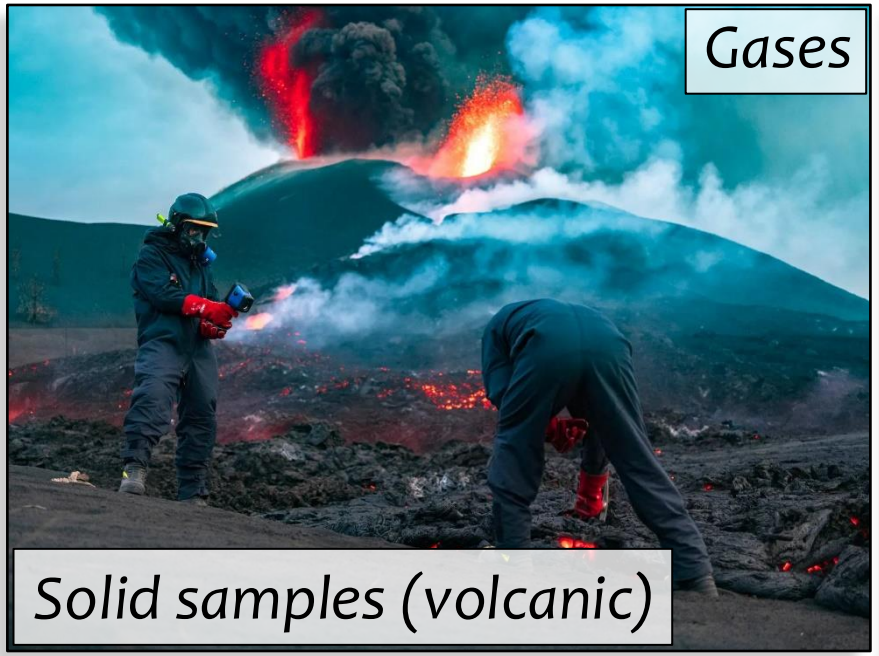
What do we want to analyse?

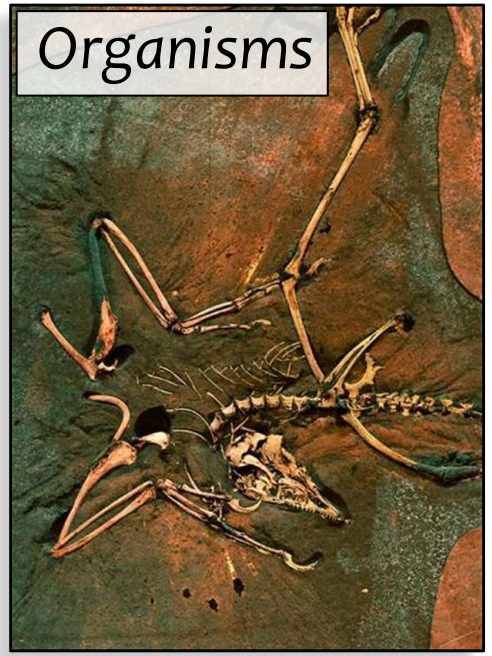












What do we want to analyse?

There are 4 major classes of ‘things’ we can analyse:

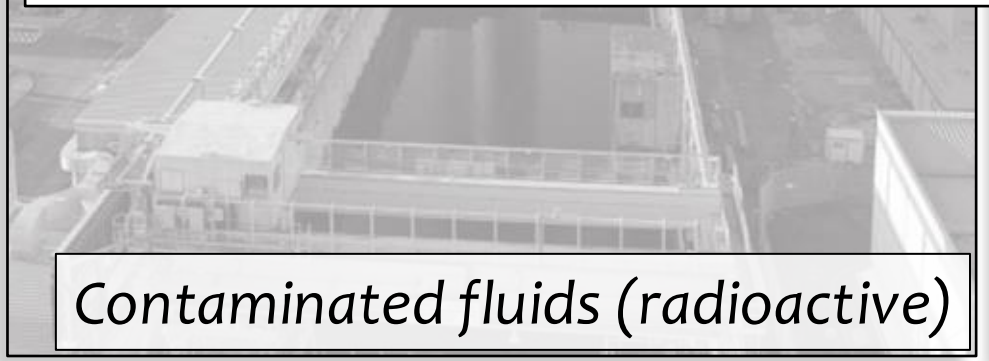
1. Bulk solid
2. Bulk liquid
3. Bulk gas
4. Interface between two phases

By the way, what do we mean by **phase**?

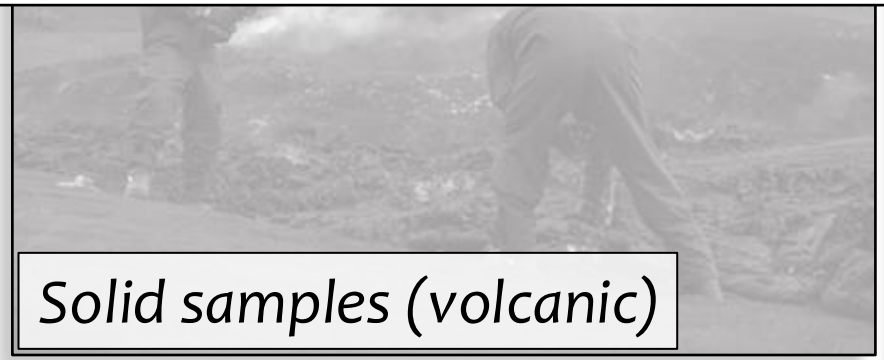
→ *region of material that is* ***chemically uniform, physically distinct,*** *and* ***mechanically separable.***



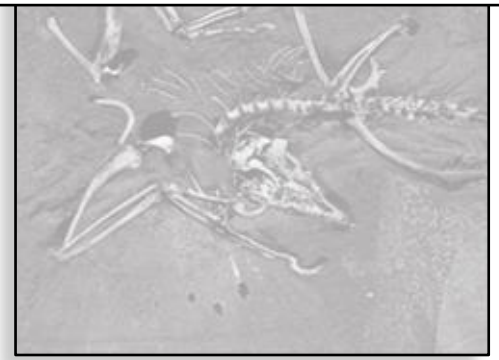
©B. Marty


Contaminated fluids (radioactive)

©Univ. Manchester


Solid samples (volcanic)

©A. Rodriguez



©K.G. Huntley/SSRL

Some terminology…

- Qualitative vs. quantitative analysis

→ *Qualitative reports list of species, while quantitative analysis reports analyte concentrations.*

- Major, minor, and trace constituents

→ *In rock or mineral analysis, major elements are those > 1% by mass (wt%), minor are between 0.1 and 1 wt%, and trace elements are <0.1 wt%.*

- Bulk vs. *in situ* analysis

→ *Bulk analyses measure overall composition of a homogeneous sample, such as a rock powder or a water sample. When information are needed on individual components of a complex mixture (e.g., minerals in a thin section), one can use spatially resolved or in situ analyses.*

- Chemical form - speciation

→ *Chemical form (e.g., oxidation state) of an element in a sample, for example sulfur may exist in a rock as sulfate (SO_4^{2-}), sulfide (S^{2-}), or elemental sulfur (S).*

- Isotopes

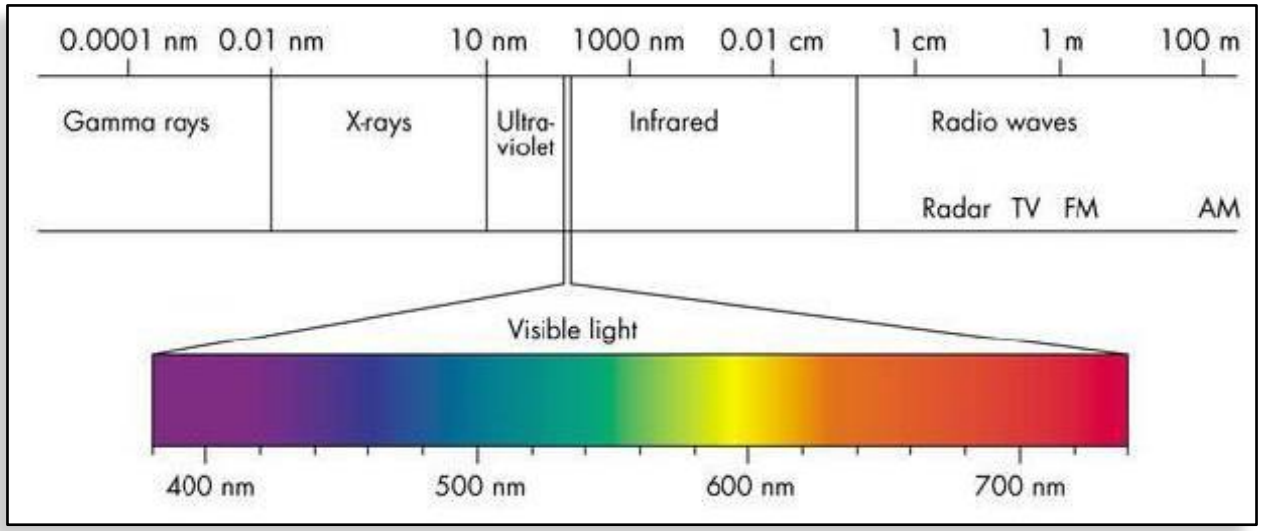
→ *Two (or more) types of atoms with the same atomic number (number of protons in nuclei), but different numbers of neutrons in their nuclei (e.g., ^{12}C, ^{13}C, ^{14}C – 6 protons and 6-8 neutrons).*

Interactions between matter and radiation source(s)

Most techniques used to characterise materials relevant to Earth and Environmental sciences involve investigating their **interactions** with **particles** (ions, protons, molecules) or some form of **electromagnetic radiation** (e.g., visible light, ultraviolet or infrared photons, X-rays, γ-rays).

This interaction may involve **diffraction**, **absorption**, **reflection**, **fluorescence**, or **scattering** of a beam of radiation.

Therefore, the response of the material investigated will largely depend on the physical and chemical properties of its constitutive elements.

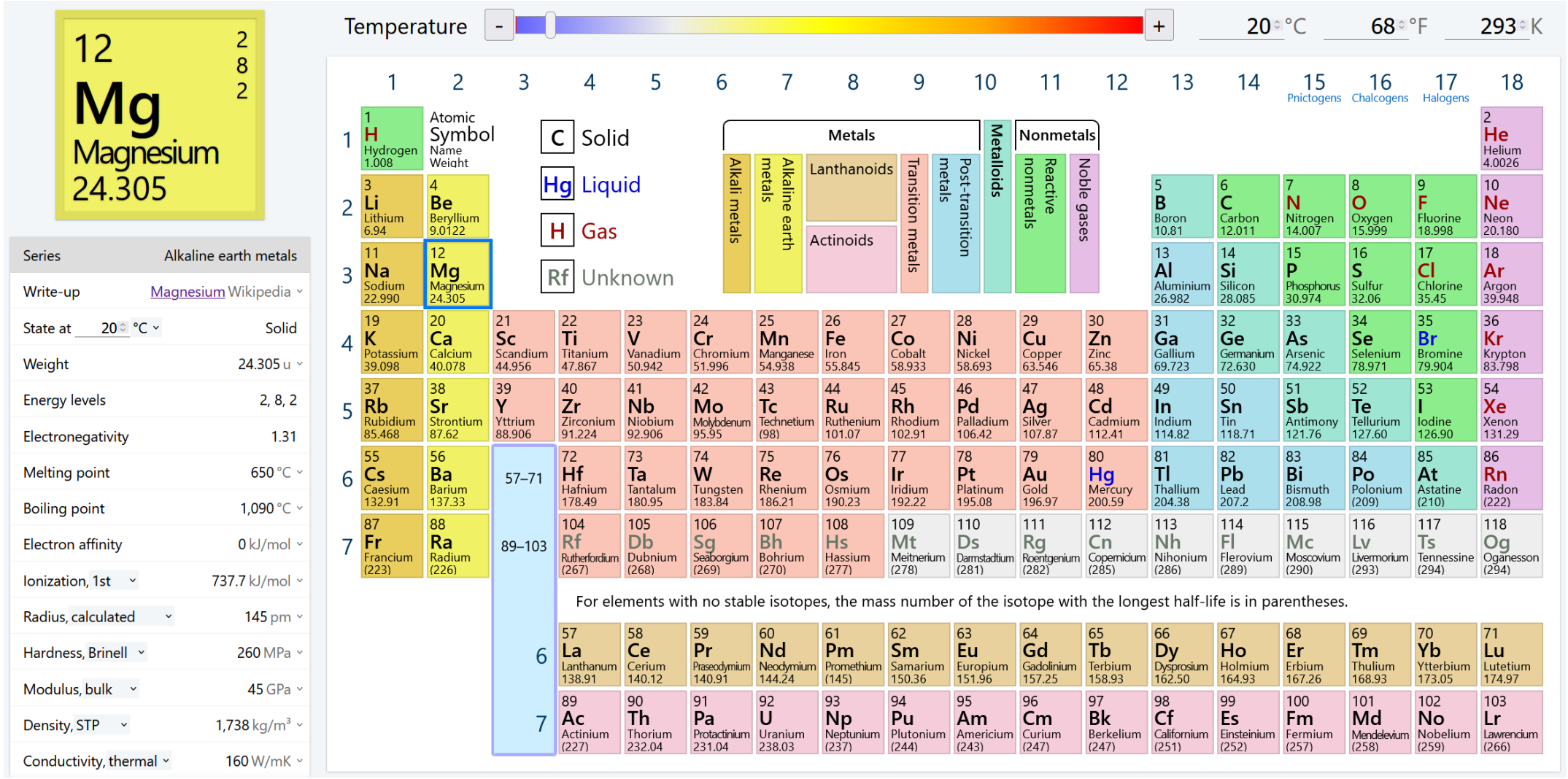


The electromagnetic spectrum

https://munsell.com/color-blog/chemistry-fireworks-colors/

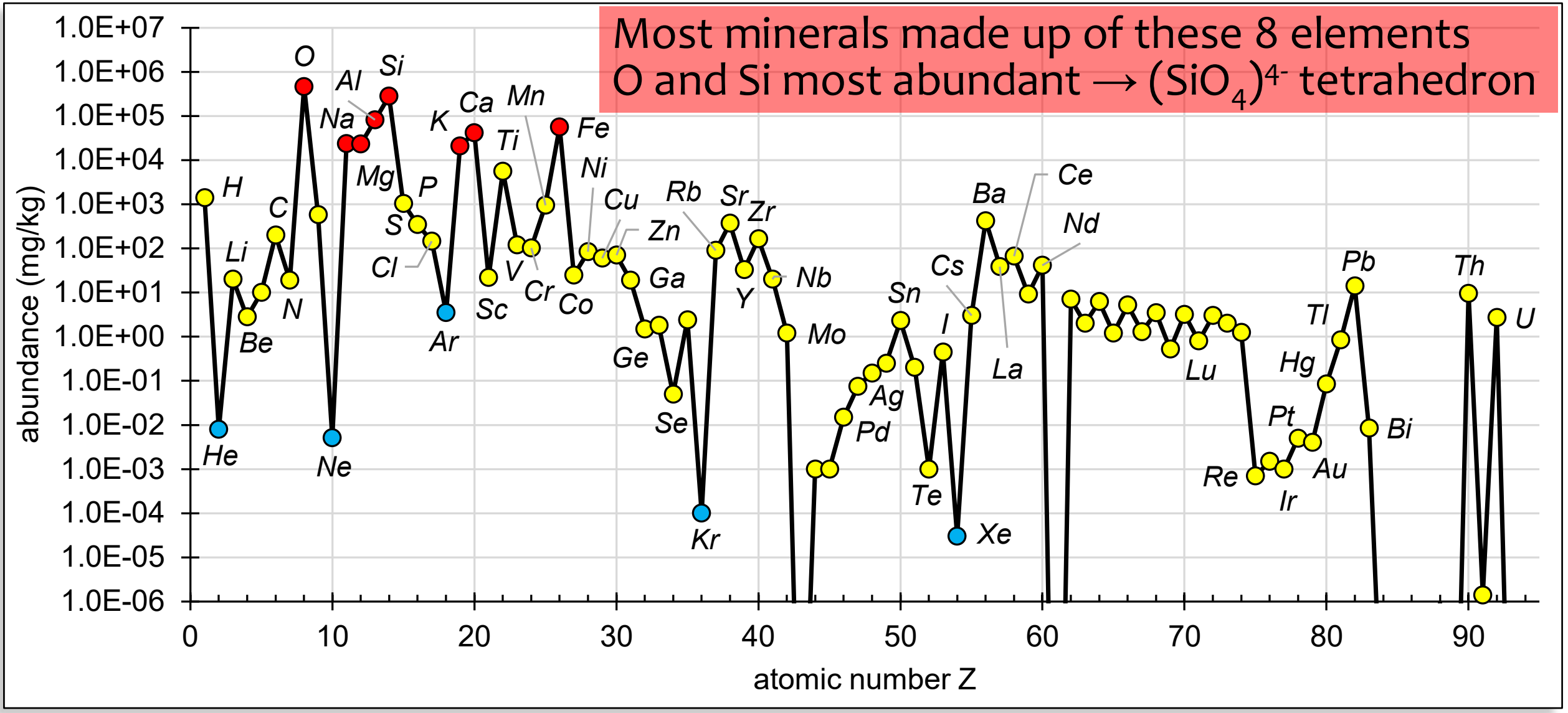
Periodic table of the elements

Each element has a unique electronic configuration, and thus unique properties.

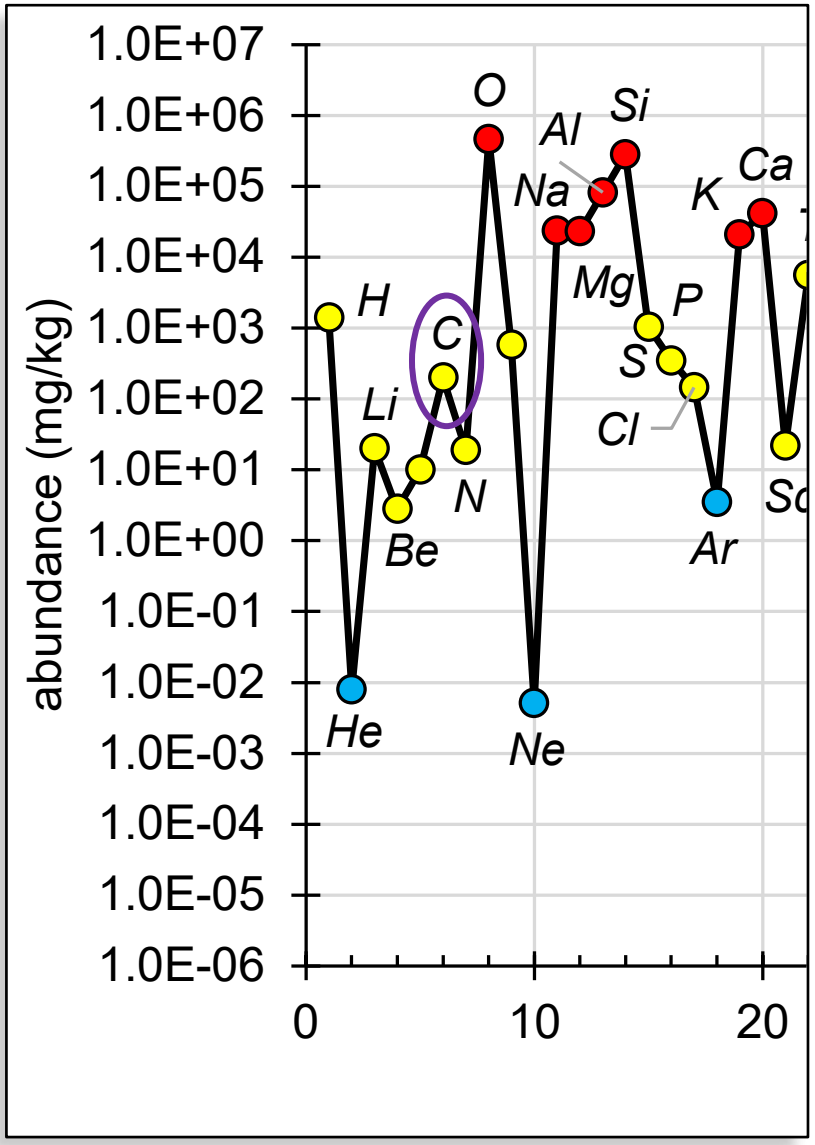


https://ptable.com/?lang=en#Properties

Element abundance in the continental crust



Element abundance in the continental crust



→ why is organic chemistry **C-based**, not Si-based?

Carbon has 4 e^- in outer shell ($[He]2s^22p^2$), and wants to gain another 4 e^- to attain max stability ($[He]2s^22p^6$) → can form 4 strong **covalent bonds**.

Carbon can form covalent bonds with many elements (e.g. CH_4), including itself (C_2H_6, C_3H_8, etc; *alkanes*) → can form **complex macromolecules**.

Carbon covalent bonds are strong, notably at temperatures relevant to life. **C–C bonds are stronger than Si–Si bonds** → C-based compounds stay together in water, while Si-based compounds break apart.

Element abundance in the continental crust

Silica-based compounds dominate the **geosphere** → we need chemical and structural information in the solid state.

Carbon-based compounds dominate the **biosphere** and are extremely diverse → again, we need chemical and structural methods.

In the **aqueous phase**, the transporting medium in many cases, there will be **C-based compounds** as well as **traces of other elements** → we need methods that can resolve trace amounts of organic and inorganic species.

Workhorse techniques in EES

- **X-ray diffraction** (XRD)
  → *Structure of solids*
- **Scanning Electron Microscope** (SEM) & **Electron Probe Micro-Analyser** (EPMA)
  → *Composition of solids*
- **Inductively Coupled Plasma** (ICP) techniques
  → *Composition of fluids and solids*
- **Gas and Liquid Chromatography** (GC-LC) techniques
  → *Composition of molecules, including proteins, peptides, pollutants, etc.*
- **GC- and LC-Mass Spectrometry** (MS) techniques
  → *Determination of molecules, Fragmentation by tandem MS*
- **FTIR & Raman spectroscopy** techniques
  → *Characterisation studies*
- **Micro-CT** techniques
  → *Determination internal structural properties, 3D imaging at micron scale*